Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/27/2021
PI Name:	Eisch, Amelia Ph.D.		
Project Title:	HZE Particle Exposure-Induced Improvement of Pattern Separation in Mature Mice: Alterations in Mission-Relevant Behaviors and Neural Circuitry		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Performance Goal No.:			
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Task Description:	ED. NOTE: New grant number 80NSSC17K0060--Continuation of "HZE Particle Exposure-Induced Improvement of Pattern Separation in Mature Mice: Alterations in Mission-Relevant Behaviors and Neural Circuitry," grant NNX15AE09G, with the same Principal Investigator (PI) Dr. Amelia Eisch; PI moved to Children's Hospital of Philadelphia/Univ Pennsylvania Perelman School of Medicine from University of Texas Southwestern Medical Center at Dallas. An unavoidable consequence of deep space missions is exposure to galactic cosmic radiation (GCR), which includes high (H) atomic number (Z) and energy (E) particles particles like Fe, Si, and O. Estimating radiation risks to the central nervous system (CNS) by HZE particles encountered during space missions is a high research priority. Past research has shown that rodents exposed to HZE particles have cognitive and performance deficits in numerous behavioral tasks, including those that rely on the hippocampus, a brain region involved in learning and memory. Notably, we have found that mature mice (of equivalent age to astronauts) exposed to either Si or Fe HZE particles actually show improved performance on a very difficult hippocampal task to assess the ability to discrimination two contexts that differ in discrete ways. Here we propose three aims to understand this improved context “pattern separation” after HZE particle exposure. In Aim 1, we hypothesize that HZE particle exposure-induced improved pattern separation is linked to improved performance on related learning and memory tasks, as well as executive function tasks, in the short-term, but to decreased performance in the long-term. In Aim 2, we hypothesize that HZE particle exposure-induced improved pattern separation is linked in the short-term to diminished stress-induced emergence of anxiety and depression-like behaviors, but to greater emergence in the long-term. In Aim 3, we hypothesize that HZE particle exposure-induced improved pattern separation is associated with disrupted hippocampal-cortical neural networks. All aims will rely on		

	both classic and cutting-edge techniques. In sum, these aims will address whether the HZE particle exposure-induced improvement in pattern separation is beneficial or detrimental to mission success (Aims 1, 2), will indicate the integrity of neural circuitry contributing to mission-relevant behaviors (Aim 3), and will define both the short- and long-term health of neural networks needed to complete deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An unavoidable aspect of manned space flights is exposure to galactic cosmic radiation (GCR), which is made up primarily of protons (87%), followed by helium (11%) and then high atomic number (Z) and high-energy (HZE) particles (2%) like iron, silicon, and oxygen. The energy of HZE particles can be very high (>1000 MeV/u), sufficient in many cases to penetrate the spacecraft hull and interior materials, and they have a complex track structure and high linear energy transfer (LET). Thus, while the greatest physical radiation dose of GCR comes from high-energy protons, the greatest biological radiation dose of GCR comes from HZE particles. With long-duration and exploratory space missions in the near future, we need to understand how GCR influences human health and behavior. Estimation of radiation risks to the central nervous system (CNS) is a high research priority according to both a National Academy of Sciences report and NASA's Radiation Health Bioastronautics Roadmap. In rodents, HZE particles induce cognitive domain deficits, including decreased hippocampal learning and memory and cortically based executive function. However, it remains unclear whether the age at irradiation (IRR) influences the outcome of behavioral tests. It is also unclear whether these cognitive decrements extend to other cognitive tests, or to other behavioral domains, such as mood and stress response. Recently, we found that mice exposed at maturity ("astronaut-aged") to 28Si or 56Fe HZE particle IRR perform better than control mice on a hippocampal-based pattern separation task (context discrimination fear conditioning, CDFC). We want to understand if this behavioral improvement in pattern separation is reflective of other changes in behavior, and whether these changes will be beneficial or detrimental to mission success. We propose a behavioral domain- and brain network-based analysis to understand the HZE particle-induced behavioral improvement shown in our pilot data. These data will have relevance for understanding the risks facing crew members in deep space missions, particularly in regards to the age of crew member at the time of the mission.
Task Progress:	January 2022: We have published that mice irradiated at "astronaut-age" show improved hippocampal-dependent performance in an aversive task (context discrimination fear conditioning, CDFC). Notably, we have found that mature mice (of equivalent age to astronauts) exposed to either 28Si or 56Fe HZE particle irradiation (IRR) actually show improved performance on a difficult hippocampal task (discrimination of 2 similar contexts) without influencing performance on an easier hippocampal task or other behaviors. For NASA Space Research Laboratory (NSRL)16B, we found that males have improved pattern separation (location discrimination, LD). This extended our findings on IRR-induced improvement in pattern separation from an aversive task (CDFC) to an appetitive task (LD). These ground-based studies on the cognitive impact of space radiation show male rodents exposed to a single high-energy particle unexpectedly have enhanced discrimination learning with similar stimuli. However, far less work has been done in female rodents. It was specifically unknown how space radiation influences cognition of female rodents when assessed via touchscreen testing, a translationally-relevant approach that reveals integrity of many cognitive domains. In mice run in NSRL17B and 18A, we showed mature female mice exposed to a mission-relevant dose of a space radiation particle have a) improved performance on a hippocampal-dependent task, b) no change in performance on a prefrontal cortex-dependent task, and c) impaired performance on a striatum-dependent behavior task. These data support the idea of competition between hippocampal and striatal memory systems, and fill a key knowledge gap on the influence of space radiation on female cognition. September 2021: Using mice irradiated at astronaut-age and multiple time points post-irradiation, we have found that mice of astronaut age show improved pattern separation in both appetitive and aversive tasks. Our ongoing work is revealing the sex- and task-dependency of this result, which is being prepared for publications. In addition to our ongoing work on how the translationome of genetically-defined dentate gyrus granule cells is altered after radiation, we are collaborating with other investigators to determine the efficacy of key countermeasures that may prevent irradiation-induced behavioral changes. This grant (NNX15AE09G/80NSSC17K0060) has funded progress on the following publications (19): Latchney SE*, Jiang Y*, Petrik DP, Eisch AJ^, Hsieh J^. Inducible knockout of Mef2A/C/D from nestin-expressing stem/progentior cells and their progeny unexpectedly uncouples neurogenesis and dendritogenesis in vivo. The Journal of the Federation of American Societies for Experimental Biology (The FASEB Journal). 29(12):5059-5071 (2015). PMID 26286136, PMC 4653059. *equal contribution ^co-corresponding authors Petrik D, Latchney SE*, Masiulis I*, Yun S*, Zhang Z*, Wu JI^, Eisch AJ^. Chromatin remodeling factor Brg1 supports the early maintenance and late responsiveness of nestin-lineage adult neural stem and progenitor cells. Stem Cells. 33(12):3655-3665 (2015). PMID 26418130, PMC 4713255. *equal contribution, listed alphabetically ^co-corresponding authors Norbury JW, Schimmerling W, Slaba TC, Azzam E, Badavi FF, Baiocco G, Benton E, Bindi V, Blakely EA, Blattnig SR, Boothman DA, Borak1 TB, Britten RA, Curtis S, Dingfelder M, Durante M, Dynan W, Eisch AJ, Elgart SR, Goodhead DT, Guida PM, Heilbronn LH, Hellweg CE, Huff JL, Kronenberg A, La Tessa C, Lowenstein D, Miller J, Morita T, Narici L, Nelson GA, Norman RB, Ohnishi T, Ottolenghi A, Patel ZS, Reitz G, Adam Rusek A, Schreurs A-S, Scott-Carnell LA, Semones E, Shay JW, Shurshakov VA, Sihver L, Simonsen LC, Story M, Turker MS, Uchihori Y, Williams J, Zeitlin CJ. Galactic cosmic ray simulation at the NASA Space Radiation Laboratory. Life Sciences in Space Research, 8:38-51 (2016). PMID 26948012, PMC 5771487. Mendoza ML, Anderson EM, Kourrich S, Eisch AJ. A NAc for Spinal Adjustments After Cocaine And Stress. Biological Psychiatry, 79(11):872–874 (2016). PMID 27198520, PMC 5784216. Yun S, Reynolds RP, Masiulis I, Eisch AJ. Re-evaluating the link between neuropsychiatric disorders and dysregulated adult neurogenesis. Nature Medicine, 22(11):1239-1247 (2016). PMID 27783068, PMC 5791154. Bulin SE, Mendoza ML, Richardson DR, Song KH, Solberg TD, Yun S, Eisch AJ. Dentate Gyrus Neurogenesis Ablation via Cranial Irradiation Enhances Morphine Self-administration and Locomotor Sensitization. Addiction Biology. 23(2):665-675 (2017). PMID 27198520, PMC 5775053. Whoolery CW, Walker AK, Lucero MJ, Richardson DR, Reynolds RP, Beddow DH, Clark KL, Shih H-Y, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Ahn F, Bulin SE, DeCarolis NA, Rivera PD, Chen BPC, Yun S, Eisch AJ. Whole Body Exposure to 28Si Radiation Dose-Dependently Disrupts Dentate Gyrus Neurogenesis and Proliferation in the Short-Term and New Neuron Survival and Contextual Fear Conditioning in the Long-Term. Radiation Research, 188(5):532-551 (2017). PMID 28945526, PMC 5901735. Escamilla (Ochoa) C*, Filonova I*, Walker AK, Xuan Z, Holehonnur R, Espinosa F, Liu S, Thyme SB, Lopez-Garcia IA, Mendoza DB, Usui N, Ellegood J, Eisch AJ, Konopka G, Lerch JP, Schier AF, Speed HE, Powell CM. Kctd13 deletion reduces synaptic transmission via increased RhoA. Nature, 9;551(7679):227-231 (2017). PMID 29088697, PMC 5787033. *equal contribution Bulin SE, Mendoza ML, Richardson DR, Song KH, Solberg TD, Yun S, Eisch AJ. Dentate Gyrus Neurogenesis Ablation via Cranial Irradiation Enhances Morphine Self-administration and Locomotor Sensitization. Addiction Biology. 23(2):665-675 (2017). PMID 27198520, PMC 5775053. Yun S, Reynolds RP, Petrof I, White A. Rivera PD, Segev A, Gibson AD, Suarez M, Desalle MJ, Ito N, Mukherjee S, Richardson DR, Kang CE, Ahrens-Nicklas RC, Soler I, Chetkovich DM, Kourrich S, Coulter DA, Eisch AJ. Stimulation of entorhinal cortex-dentate gyrus

circuitry is antidepressive. Nature Medicine, 24(5):658-666 (2018). PMID 29662202, PMC 5948139.

Zanni G*, Deutsch HM*, Rivera PD, Shih H-Y, LeBlanc JA, Amaral WZ, Lucero MJ, Redfield RL, DeSalle MJ, Chen BPC, Whoolery CW, Reynolds RP, Yun S, Eisch AJ. Whole-body 12C irradiation transiently decreases mouse hippocampal dentate gyrus proliferation and immature neuron number, but does not change new neuron survival rate. International Journal of Molecular Sciences, 19(10), pii:E3078. doi: 10.3390/ijms19103078 (2018). PMID 30304778, PMC 6213859. *equal contribution Rivera PD, Simmons SJ, Reynolds RP, Just AL, Birnbaum SG, Eisch AJ. Image-guided cranial irradiation-induced ablation of dentate gyrus neurogenesis impairs extinction of recent morphine reward memories. Hippocampus. Feb 18. doi: 10.1002/hipo.23071 (2019). PMID 30779299, PMC 7036142.

Snitow ME, Zanni G, Ciesielski B, Burgess-Jones P, Eisch AJ, O'Brien WT, Klein PS. Adult hippocampal neurogenesis is not necessary for the response to lithium in the forced swim test. Neuroscience. Mar 30; 704: 67-72. doi: 10.1016/j.neulet.2019.03.052 (2019). PMID 30940476, PMC 6594907.

Bulin SE*, Simmons SJ*, Richardson DR, Latchney SE, Deutsch HM, Yun S, Eisch AJ. Indices of dentate gyrus neurogenesis are unaffected immediately after or following withdrawal from morphine self-administration compared to saline self-administering control male rats. Behavioural Brain Research. Mar 2, 381: 112448. doi: 10.1016/j.bbr.2019.112448, Epub Dec 20 (2019). PMID 31870778, PMC 7036141. *equal contribution

Whoolery CW*, Yun S*, Reynolds RP, Lucero MJ, Soler I, Tran FH, Ito N, Redfield RL, Richardson DR, Shih H-Y, Rivera PD, Chen BPC, Birnbaum SG, Stowe AM, Eisch AJ. Multi-domain cognitive assessment of male mice shows space radiation is not harmful to high-level cognition and actually improves pattern separation. Scientific Reports. Feb 17, 10(1):2737. doi: 10.1038/s41598-020-59419-z (2020). PMID 32066765, PMC 7026431 *equal contribution

Ortega SB*, Torres VO*, Latchney SE, Whoolery CW, Noorbhai IZ, Poinsatte K, Selvaraj UM, Benson MA, Meeuwissen AJM, Plautz EJ, Kong X, Ramirez DM, Ajay AD, Meeks JP, Goldberg MP, Monson NL, Eisch AJ, Stowe AM. B cells migrate into remote brain areas and support neurogenesis and functional recovery after focal stroke in mice. Proceedings of the National Academy of Sciences USA. Feb 12:201913292. doi: 10.1073/pnas.1913292117. Epub ahead of print. (2020) PMID 32051245, PMC 7060723. *equal contribution

Tran FH*, Spears SL*, Ahn KJ, Eisch AJ, Yun S. Does chronic systemic injection of the DREADD agonists clozapine-N-oxide or compound 21 change behavior relevant to locomotion, exploration, anxiety, and depression in male non-DREADD-expressing mice? (2020). Neuroscience Letters. doi: 10.1016/j.neulet.2020.135432. PMID 33080350. *equal contribution.

Clark LR, Yun S, Acquah NK, Kumar PL, Metheny HE, Paixao RCC, Cohen AS, Eisch AJ. Mild traumatic brain injury induces transient, sequential increases in proliferation, neurogenesis, and cell survival: a time course study in the male mouse dentate gyrus. Frontiers in Neuroscience special issue "New Insights into Adult Neurogenesis and Neurodegeneration: Challenges for Brain Repair". Jan 7:2021 (14)1372. doi: org/10.1101/2020.10.07.330118. (2021) <a target="_blank" href="http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=pubmed&dopt=Abstract&list_uids=33488351">PMID:33488351</

Soler I^, Yun S^*, Reynolds RP^^, Whoolery CW^^, Tran FH, Kumar PL, Rong Y, DeSalle MJ, Gibson AD, Stowe AM, Kiffer FC, Eisch AJ*. Multi-Domain Touchscreen-Based Cognitive Assessment of C57BL/6J Female Mice Shows Whole-Body Exposure to 56Fe Particle Space Radiation in Maturity Improves Discrimination Learning Yet Impairs Stimulus-Response Rule-Based Habit Learning. Accepted, Frontiers in Behavioral Neuroscience. ^equal contribution co-first authors, ^^equal contribution co-second authors, *co-corresponding authors

Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Soler I, Yun S, Reynolds RP, Whoolery CW, Tran FH, Kumar PL, Rong Y, DeSalle MJ, Gibson AD, Stowe AM, Kiffer FC, Eisch AJ. "Multi-domain touchscreen-based cognitive assessment of C57BL/6J female mice shows whole-body exposure to 56Fe particle space radiation in maturity improves discrimination learning yet impairs stimulus-response rule-based habit learning." Front Behav Neurosci. 2021 Oct 11;15:722780. https://doi.org/10.3389/fnbeh.2021.722780 ; PMID: 34707486; PMCID: PMC8543003 , Oct-2021
Articles in Peer-reviewed Journals	Clark LR, Yun S, Acquah NK, Kumar PL, Metheny HE, Paixao RCC, Cohen AS, Eisch AJ. "Mild traumatic brain injury induces transient, sequential increases in proliferation, neuroblasts/immature neurons, and cell survival: A time course study in the male mouse dentate gyrus." Front Neurosci. 2021 Jan 7;14:612749. https://doi.org/10.3389/fnins.2020.612749 ; PMID: 33488351; PMCID: PMC7817782 , Jan-2021
Articles in Peer-reviewed Journals	Kiffer FC, Luitel K, Tran FH, Patel RA, Guzman CS, Soler I, Xiao R, Shay JW, Yun S, Eisch AJ. "Effects of a 33-ion sequential beam galactic cosmic ray analog on male mouse behavior and evaluation of CDDO-EA as a radiation countermeasure." Behav Brain Res. 2022 Feb 15;419:113677. https://doi.org/10.1016/j.bbr.2021.113677 ; PMID: 34818568; PMCID: PMC9755463 , Feb-2022